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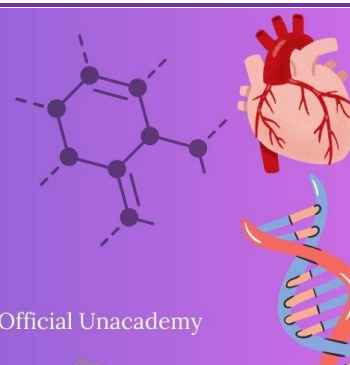
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LAST 10 YEARS

NEET PYQ



Chapter 8 : Cell : The Unit of Life

1. Given below are two statements : one is Assertion (A), and the other is Reason (R). [NEET-2025]

Assertion (A) : The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R) : Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both A and R are true and R is the correct explanation of A
- (2) Both A and R are true but R is not the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

2. From the statements given below choose the correct option : [NEET-2025]

- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
- B. Each ribosome has two sub-units.
- C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
- D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
- E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S.

- (1) A, B, C are true
- (2) A, B, D are true
- (3) A, B, E are true
- (4) B, D, E are true

3. A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is [NEET-2025]

- | | |
|--------------|---------------------------|
| (1) Mesosome | (2) Chromatophores |
| (3) Cristae | (4) Endoplasmic Reticulum |

4. Match List-I with List- II.

[NEET-2025]

List-I	List-II
A. Centromere	I. Mitochondrion
B. Cilium	II. Cell division
C. Cristae	III. Cell movement
D. Cell membrane	IV. Phospholipid Bilayer
(1) A-I, B-II, C-III, D-IV	(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-II, C-III, D-I	(4) A-II, B-III, C-I, D-IV

5. Match List-I with List-II:

[Re-NEET-2024]

List-I	List-II
A. Fleming	I. Disc shaped sacs or cisternae near cell nucleus
B. Robert Brown	II. Chromatin
C. George Palade	III. Ribosomes
D. Camillo Golgi	IV. Nucleus
(1) A-II, B-IV, C-III, D-I	(2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-III, D-IV	(4) A-IV, B-II, C-III, D-I

6. Mesosome in a cell is a :

[Re-NEET-2024]

- (1) Membrane bound vesicular structure
- (2) Chain of many ribosomes attached to a single mRNA
- (3) Special structure formed by extension of plasma membrane
- (4) Medium sized chromosome

7. Match List-I with List-II :

[Re-NEET-2024]

List-I	List-II
A. Metacentric chromosome	I. Chromosome has a terminal centromere
B. Sub-metacentric chromosome	II. Middle centromere forming two equal arms of chromosome
C. Acrocentric chromosome	III. Centromere is slightly away from the middle of chromosome resulting into two unequal arms
D. Telocentric chromosome	IV. Centromere is situated close to its end forming one extremely short and one very long arm
(1) A-II, B-I, C-IV, D-III	(2) A-IV, B-I, C-II, D-III
(3) A-I, B-II, C-III, D-IV	(4) A-II, B-III, C-IV, D-I

8. Match List-I with List-II. :

[Re-NEET-2024]

List-I	List-II
A. F1 Particles	I. Chromosomes
B. Histones	II. Cilia
C. Axoneme	III. Golgi apparatus
D. Cisternae	IV. Mitochondria

Choose the correct answer from the options given below:

(1) A-II, B-I, C-IV, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-IV, B-III, C-I, D-II

9. Match List I with List II :

[NEET-2024]

List I	List II
A. Axoneme	I. Centriole
B. Cartwheel pattern	II. Cilia and flagella
C. Crista	III. Chromosome
D. Satellite	IV. Mitochondria

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-II, C-III, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

10. The DNA present in chloroplast is:

[NEET-2024]

(1) Linear, double stranded

(2) Circular, double stranded

(3) Linear, single stranded

(4) Circular, single stranded

11. Given below are two statements:

[NEET-2024]

Statement I: Concentrically arranged cisternae of Golgi complex are arranged near the nucleus with distinct convex cis or maturing and concave trans or forming face.

Statement II: A number of proteins are modified in the cisternae of Golgi complex before they are released from cis face.

In the light of the above statements, choose the correct answer from the option given below.

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

12. Match List I with List II :

[NEET-2024]

List-I	List-II
A. Nucleolus	I. Site of formation of glycolipid
B. Centriole	II. Organization like the cartwheel
C. Leucoplasts	III. Site for active ribosomal RNA synthesis
D. Golgi apparatus	IV. For storing nutrients

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-I, B-II, C-III, D-IV

13. Given below are two statements:

Statement I: Mitochondria and chloroplasts both double membranes bound organelles.

Statement II: Inner membrane of mitochondria is relatively less permeable, as compared chloroplast.

In the light of the above statements, choose the mis appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

14. Which of the following statements are correct with respect to Golgi apparatus? [NEET-2023-MANIPUR]

- (A) It is the important site of formation of glycoprotein and glycolipids.
- (B) It produces cellular energy in the form of ATP.
- (C) It modifies the protein synthesized by ribosomes on ER.
- (D) It facilitates the transport of ions.
- (E) It provides mechanical support.

Choose the most appropriate answer from the options given below :

- (1) (A) and (D) only
- (2) (D) and (E) only
- (3) (B) and (C) only
- (4) (A) and (C) only

15. Given below are two statements:

[NEET-2023-MANIPUR]

Statement I : In bacteria, the mesosomes are formed by the extensions of plasma membrane.

Statement II: The mesosomes, in bacteria, help in DNA replication and cell wall formation.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are correct
- (2) Both Statement I and Statement II are incorrect
- (3) Statement I is correct but Statement II is incorrect
- (4) Statement I is incorrect but Statement II is correct

16. How many different proteins does the ribosome consist of?

[NEET-2023]

- (1) 80
- (2) 60
- (3) 40
- (4) 20

17. Which of the following are NOT considered as the part of endomembrane system?

[NEET-2023]

- A. Mitochondria
- B. Endoplasmic reticulum
- C. Chloroplasts
- D. Golgi complex
- E. Peroxisomes

Choose the most appropriate answer from the options given below:

- (1) B and D only
- (2) A, C and E only
- (3) A and D only
- (4) A, D and E only

18. Which of the following functions is carried out by cytoskeleton in a cell?

[NEET-2023]

- (1) Nuclear division
- (2) Protein synthesis
- (3) Motility
- (4) Transportation

19. Match List-I with List-II.

[NEET (UG) 2022]

List-I

List-II

- | | |
|----------------------------|---|
| (a) Metacentric | (i) Centromere situated close to the end forming one extremely short and one very long arms |
| (b) Acrocentric chromosome | (ii) Centromere at the terminal end |
| (c) Sub metacentric | (iii) Centromere in the middle forming two equal arms of chromosomes |
| (d) Telocentric chromosome | (iv) Centromere slightly away from the middle forming one shorter arm and one longer arm |

(1) (a)-(i),(b)-(iii),(c)-(ii),(d)-(iv)

(2) (a)-(ii),(b)-(iii),(c)-(iv),(d)-(i)

(3) (a)-(i),(b)-(ii),(c)-(iii),(d)-(iv)

(4) (a)-(iii),(b)-(i),(c)-(iv),(d)-(ii)

20. Which of the following statements with respect to Endoplasmic Reticulum is incorrect? [NEET (UG) 2022]

- (1) SER is devoid of ribosomes
- (2) In prokaryotes only RER are present
- (3) SER are the sites for lipid synthesis
- (4) RER has ribosomes attached to ER

21. Which type of substance would face difficulty to pass through the cell membrane? [Re-NEET (UG) 2022]

- (1) Substance with hydrophobic moiety
- (2) Substance with hydrophilic moiety
- (3) All substance irrespective of hydrophobic and hydrophilic moiety
- (4) Substance soluble in lipids

22. If the pH in lysosomes is increased to alkaline, what will be the outcome? [Re-NEET (UG) 2022]

- (1) Hydrolytic enzymes will function more efficiently
- (2) Hydrolytic enzymes will become inactive
- (3) Lysosomal enzymes will be released into the cytoplasm
- (4) Lysosomal enzymes will be more active

23. The organelles that are included in the endomembrane system are: [NEET (UG) 2021]

- (1) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes
- (2) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles
- (3) Golgi complex, Mitochondria, Ribosomes and Lysosomes
- (4) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes

24. When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as: [NEET (UG) 2021]

- | | |
|---------------------|-----------------|
| (1) Metacentric | (2) Telocentric |
| (3) Sub-metacentric | (4) Acrocentric |

25. Match List -I with List - II. [NEET (UG) 2021]

List -I

List -II

- | | |
|----------------|--|
| (a) Cristae | (i) Primary constriction in chromosome |
| (b) Thylakoids | (ii) Disc-shaped sacs in Golgi apparatus |
| (c) Centromere | (iii) Infoldings in mitochondria |
| (d) Cisternae | (iv) Flattened membranous sacs in stroma of plastids |

Choose the correct answer from the options given below.

- | | |
|--|--|
| (1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i) | (2) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii) |
| (3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii) | (4) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i) |

26. Which of the following is an incorrect statement? [NEET (UG) 2021]
- (1) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles.
 - (2) Microbodies are present both in plant and animal cells.
 - (3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm.
 - (4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm.
27. Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells [NEET - 2020]
- (1) Polysomes
 - (2) Endoplasmic reticulum
 - (3) Peroxisomes
 - (4) Golgi bodies
28. The shorter and longer arms of a submetacentric chromosome are referred to as :- [NEET-UG 2019]
- (1) s-arm and l-arm respectively
 - (2) p-arm and q-arm respectively
 - (3) q-arm and p-arm respectively
 - (4) m-arm and n-arm respectively
29. Which of the following pair of organelles does not contain DNA? [NEET-UG 2019]
- (1) Mitochondria and Lysosomes
 - (2) Chloroplast and Vacuoles
 - (3) Lysosomes and Vacuoles
 - (4) Nuclear envelope and Mitochondria
30. Which of the following statements is not correct? [NEET-UG 2019]
- (1) Lysosomes have numerous hydrolytic enzymes.
 - (2) The hydrolytic enzymes of lysosomes are active under acidic pH.
 - (3) Lysosomes are membrane bound structures.
 - (4) Lysosomes are formed by the process of packaging in the endoplasmic reticulum.
31. The concept of "Omnis cellula-e cellula" regarding cell division was first proposed by: [NEET-UG 2019]
- (1) Rudolf Virchow
 - (2) Theodore Schwann
 - (3) Schleiden
 - (4) Aristotle
32. Which of the following statements regarding mitochondria is incorrect? [NEET-UG 2019]
- (1) Outer membrane is permeable to monomers of carbohydrates, fats and proteins.
 - (2) Enzymes of electron transport are embedded in outer membrane.
 - (3) Inner membrane is convoluted with infoldings.
 - (4) Mitochondrial matrix contains single circular DNA molecule and ribosomes.
33. Which of the following cell organelles is present in the highest number in secretory cells? [NEET- 2019 (Odisha)]
- (1) Mitochondria
 - (2) Golgi complex
 - (3) Endoplasmic reticulum
 - (4) Lysosomes

34. Non-membranous nucleoplasmic structures in nucleus are the site for active synthesis of:-

- | | | |
|-----------------------|----------|-----------------------|
| (1) Protein synthesis | (2) mRNA | [NEET- 2019 (Odisha)] |
| (3) rRNA | (4) tRNA | |

35. Match the column-I with column-II :-

[NEET- 2019 (Odisha)]

- | Column-I | Column-II |
|---------------|--|
| (a) Golgi | (i) Synthesis of apparatus protein |
| (b) Lysosomes | (ii) Trap waste and excretory products |
| (c) Vacuoles | (iii) Formation of glycoproteins and glycolipids |
| (d) Ribosomes | (iv) Digesting biomolecules |
- (1) (a)–(iii), (b)–(iv), (c)–(ii), (d)–(i)
(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)
(3) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)
(4) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)

36. Which of the following is true for nucleolus?

[NEET-UG 2018]

- (1) Larger nucleoli are present in dividing cells.
(2) It is a membrane-bound structure.
(3) It takes part in spindle formation.
(4) It is a site for active ribosomal RNA synthesis.

37. The Golgi complex participates in

[NEET-UG 2018]

- (1) Fatty acid breakdown
(2) Formation of secretory vesicles
(3) Respiration in bacteria
(4) Activation of amino acid

38. Which of the following events does not occur in rough endoplasmic reticulum?

[NEET-UG 2018]

- | | |
|--------------------------------|----------------------------|
| (1) Protein folding | (2) Protein glycosylation |
| (3) Cleavage of signal peptide | (4) Phospholipid synthesis |

39. Select the incorrect match :

[NEET-UG 2018]

- (1) Lampbrush – Diplotene chromosomes bivalents
(2) Allosomes – Sex chromosomes
(3) Submetacentric – L-shaped chromosomes chromosomes
(4) Polytenes – Oocytes of chromosomes amphibians

40. Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?

(1) Ribosome

[NEET-UG 2017]

(2) Chloroplast

(3) Mitochondrion

(4) Lysosome

41. Mitochondria and chloroplast are :-

[NEET- I 2016]

(a) semi-autonomous organelles

(b) formed by division of pre-existing organelles and they contain DNA but lack protein synthesizing machinery

Which one of the following options is correct?

(1) Both (a) and (b) are correct

(2) (b) is true but (a) is false

(3) (a) is true but (b) is false

(4) Both (a) and (b) are false

42. Microtubules are the constituents of :-

[NEET- I 2016]

(1) Cilia, Flagella and Peroxisomes

(2) Spindle fibres, Centrioles and Cilia

(3) Centrioles, Spindle fibres and Chromatin

(4) Centrosome, Nucleosome and Centrioles

43. A complex of ribosomes attached to a single strand of RNA is known as :-

[NEET- I 2016]

(1) Polysome

(2) Polymer

(3) Polypeptide

(4) Okazaki fragment

44. Which one of the following cell organelles is enclosed by a single membrane?

[NEET- I 2016]

(1) Mitochondria

(2) Chloroplasts

(3) Lysosomes

(4) Nuclei

45. Water soluble pigments found in plant cell vacuoles are :-

[NEET- I 2016]

(1) Xanthophylls

(2) Chlorophylls

(3) Carotenoids

(4) Anthocyanins

46. A cell organelle containing hydrolytic enzymes is:-

[NEET- II 2016]

(1) Ribosome

(2) Mesosome

(3) Lysosome

(4) Microsome

ANSWER KEY

1. (2)
2. (1)
3. (1)
4. (4)
5. (1)
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9. (4)
10. (2)
11. (4)
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45. (4)
46. (3)